Vision-language models like CLIP demonstrate strong zero-shot capabilities, but adapting them to downstream tasks with prompt learning methods introduces substantial trainable parameters that limit deployment in resource-constrained scenarios. Existing multi-modal prompt learning methods employ independent projection layers per transformer layer, resulting in parameter redundancy without explicit gradient sharing across modalities and layers. We propose MMRL++, which introduces Shared-Residual Representation Aligner (SRRA) using shared base weights with layer-specific low-rank residuals, and Progressive Representation Composition (PRC) for inter-layer semantic flow, achieving parameter efficiency while maintaining multi-modal alignment. MMRL++ reduces trainable parameters by over 80% compared to MMRL while achieving competitive or superior performance across three evaluation settings: base-to-novel generalization, few-shot learning, and cross-dataset transfer on 15 datasets. Our shared-residual design enables efficient vision-language adaptation suitable for edge deployment and resource-constrained applications, while the progressive composition mechanism enhances intra-modal interaction beyond single-layer token injection.